MILITARY SPECIFICATION

IMPRESSION MATERIAL, DENTAL,

DUPLICATING, HYDROCOLLOID, AGAR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a reusable, hydrocolloid-agar impression material (duplicating material) suitable for use in the fluid-resin technique.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material. The impression material (duplicating material) shall consist of an aqueous gel containing the following percentages by formula weight of ingredients with a plus or minus 5 percent (%) tolerance on each ingredient:

Agar-agar content	4.0%
Glycerine content	52.5%
Defoaming and preservative agents content (ammonium acetate, polyethylene glycol monolaurate and methyl parahydroxybenzoate)	1.5%
Water content	42.0%

3.2 Description. The impression material shall be a thermoreversible hydrocolloid gel suitable for use in the prosthetic dental laboratory for implementation of the fluid-resin technique of denture fabrication. The impression material shall be capable of taking an accurate impression of the master cast and shall serve as an investing and duplicating medium in the fabrication of a resin denture by the fluid resin technique. The material shall be supplied in gelled form and shall be translucent and capable of being easily cut. The material shall be tacky for retaining teeth and shall make a firm seal around the necks of teeth (see 6.3).

3.3 Compatibility with fluid resin. The fluid resin (see 6.1 and 6.3 poured against the duplicating material shall produce a line 0.050 millimeters wide, and the fluid resin itself shall be capable of reproducing a line 0.025 millimeters wide, when tested as specified in 4.4.1. Upon completion of the curing cycle, the release of the cured fluid resin shall be clean, shall be smooth and shall capture all detail of the mold.

3.4 Melting time. A one-inch cube of the gelled material shall completely melt in 8 1/2 minutes when tested as specified in 4.4.2.

3.5 Gelation temperature. The melted material shall gel at 110 degrees Fahrenheit (° F.) ± 5° F. when tested as specified in 4.4.3.

3.6 Mold growth. The impression material shall not show mold growth when tested as specified in 4.4.4.

3.7 Gel strength. An 8-pound weight shall not completely penetrate a 15-millimeter (mm.) thickness of gelled material, when tested as specified in 4.4.5.

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3.8 Certificates of reuse. The material shall be reusable for a minimum of 15 times, when tested as specified in 4.4.6. The contractor shall submit to the procuring activity, a copy of the original certification, that the lots or batches offered to the Government were composed of the same formula and manufactured identically as the production lot or batch originally tested for reuse.

3.9 Volume of material. The volume of the impression material furnished in the gelled form shall be 1 gallon minimum (231 cubic inches minimum), when tested as specified in 4.4.7.

3.10 Stability. The impression material shall conform to the requirements of this specification for no less than two years from date of delivery to the Government, when stored in a cool dry place (59° F. to 86° F.).

3.11 Delivery schedule. No more than three months shall have elapsed from date of manufacture of the impression material to date of delivery to the Government.

3.12 Manufacturer's instructions. Adequate and accurate instructions for specific uses and technique for use of the impression material shall be furnished with each unit of issue. In addition, two copies of the instructions shall be forwarded to the contracting officer for information purposes.

3.13 Workmanship. Workmanship shall be first class throughout. The product and its containers shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, specification requirements, the name of the manufacturer or supplier, the FSN and the Item Identification.

4.1.4 Manufacturer's records. The percentage of agar-agar, glycerine, and defoaming and preservative agents, shall be ascertained from the manufacturer's production control records.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in 4.2.1 and table I (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 Sampling for tests. The unit of product for testing shall be one unit container taken at random from each lot or batch number. Where feasible, the same test unit may be used for the determination of two or more different test characteristics. The number of tests and determinations for each characteristic shall be as listed in table I.

Table I. Tests

Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations
Compatibility with fluid resin	3.3	4.4.1	1	1 1/	1
Melting time	3.4	4.4.2	1	2 1/	2
Gelation temperature	3.5	4.4.3	1	3 2/	3
Mold growth	3.6	4.4.4	1	1 1/	1
Gel strength	3.7	4.4.5	1	2 1/	2
Volume of material	3.9	4.4.7	1	2 2/	2

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1/ A lot shall be rejected if any determination fails the test requirement.

2/ A lot shall be rejected if the average of the determinations per unit of product fails the test.

4.3 Examination. The impression material shall be examined to determine compliance with specification requirements.

4.4 Tests.

4.4.1 Compatibility with fluid resin. A metal ring, 30 mm. inside diameter and 16 mm. high, shall be positioned on a stainless steel test block similar to that shown in figure 1A so that the intersection of a crossline and a groove 0.050 mm. wide is in the center of the ring. As an alternate, a metal ring, 30 mm. inside diameter and 16 mm. high, shall be positioned on a stainless steel test block similar to that shown in figure 1B so that the 0.050 mm. line of the test block is centered in the ring. The ring shall be slightly over-filled with the duplicating material which has been stored and poured at 125° F. ± 4° F. A flat plate shall be placed on top and the excess material shall be squeezed out. After gelation, the ring with the impression material shall be separated from the plate and the test block. The impression material shall not be separated from the ring. The ring shall be walled with Kraft paper to a height of approximately 60 mm., providing a reservoir on the side of the duplicating material in which the 0.050 mm. line is showing. Prior to pouring the fluid resin, metal parts coming in contact with the resin shall be treated with a silicone release agent. The fluid resin (see 6.1 and 6.3), shall be poured against the duplicating material. The fluid resin, prepared and cured according to the manufacturer's instructions, shall be removed from the duplicating material and examined. The reproduction of the 0.050 mm. line shall be continuous for the full width of the ring. The fluid resin itself, when allowed to cure in the ring directly against the block (figure 1A or 1B) shall reproduce the line 0.025 mm. wide, continuous for the full width of the ring.

4.4.2 Melting time. A double boiler shall be employed, using water in lower part. Water shall be brought to a boil. A one-inch cube of gelled material shall be placed in upper part of uncovered double boiler. The material shall be stirred occasionally during melting period of 8-1/2 minutes. The material shall be examined to determine that it has completely changed to the liquid state. Two tests shall be conducted.

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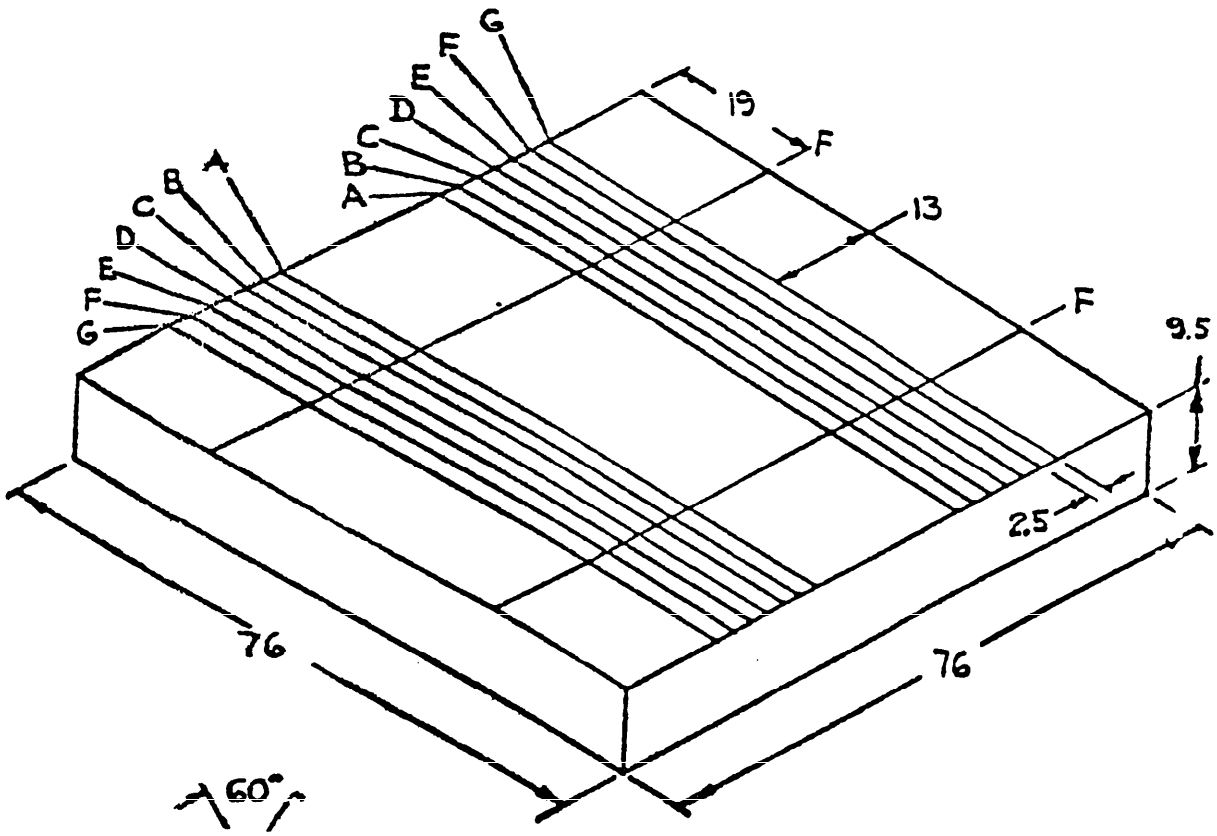
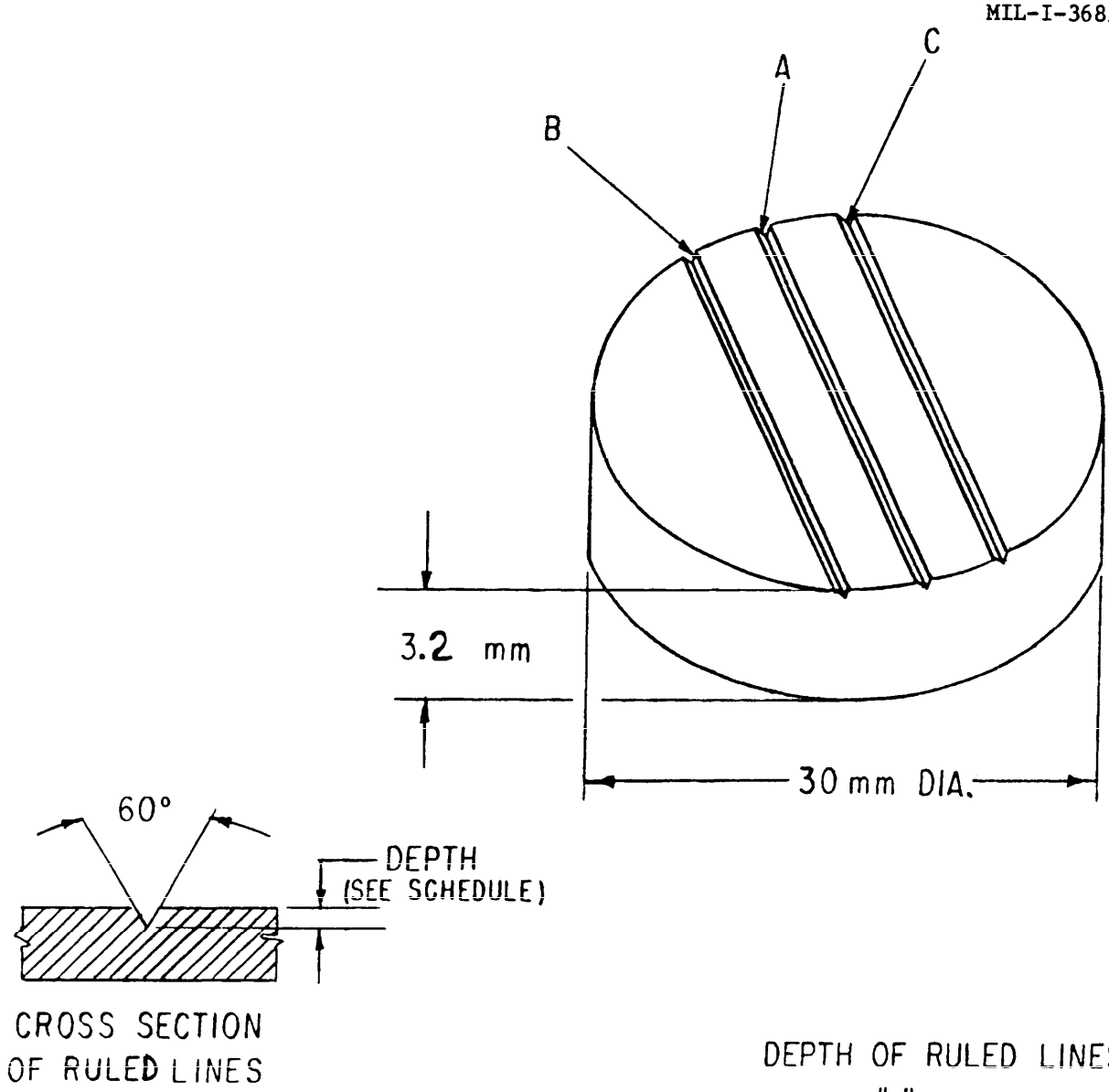


Figure 1A. Test Block for Detail Reproduction

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DEPTH OF RULED LINES:
LINE "A" – 0.025 mm
LINE "B" – 0.050 mm
LINE "C" – 0.075 mm

FIGURE 1B

ALTERNATE TEST BLOCK FOR DETAIL REPRODUCTION

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4.4.3 Gelation temperature. Approximately 50 ml. of the material at pouring temperature of 125 ± 4° F., shall be poured into a small metal tray (figure 2). As the material cools, the metal tube (figure 2) shall be inserted vertically into the material until the tube touches the floor of the metal tray. The tube shall be withdrawn immediately. Near the critical point trials shall be made every 1° F. The final trials shall be made in that portion of the material lying near the thermometer bulb. The temperature of gel formation shall be the highest temperature at which the two concentric circles, caused by the inside and outside surfaces of the identifying tubes are clearly outlined and the material does not cling to the polished surface of the tube. The rate of cooling of the specimen from 125 ± 4° F. to the temperature of gel formation shall be ± 1° F. per minute. An average of 3 tests shall be reported to the nearest degree.

4.4.4 Mold growth. Approximately 25 ml. of the material at pouring temperature of 125 ± 4° F. shall be placed in a petri dish or similar container and cooled to 73 ± 3° F. Mold obtained form stale bread or similar substance shall be used to inoculate the impression material with a needle which shall have been previously drawn through the bread mold. The material shall be maintained at 73 ± 3° F. and 95 ± 5 percent relative humidity for one week. At the end of the one week period, the material shall be examined for mold growth.

4.4.5 Gel strength. The material at pouring temperature of 125 ± 4° F., shall be poured into and shall completely fill a petri dish 50 mm. in diameter and 15 mm. in depth. After gelation, the material shall be maintained at 73 ± 3° F. and 95 ± 5 percent relative humidity for 24 hours. The dish shall be placed directly under a weight of 8 pounds ± 2 ounces, having a flat steel tip of 1 square cm. The steel tip shall be placed on the gelled material, and the weight released for 15 seconds. Penetration of steel tip into gelled material shall be observed. Two tests shall be conducted.

4.4.6 Re-use. Approximately 3 cubic inches of the gelled impression material shall be melted (4.4.2), poured at 125 ± 4° F., and gelled (4.4.3). Above cycle shall be repeated 15 times, and the material shall then meet the compatibility test of 4.4.1 (see 3.8).

4.4.7 Volume of material. The volume of the gelled impression material furnished in a unit container (see 4.2.2) shall be determined mathematically, or the material shall be melted and the volume of the liquid shall be determined using a suitable measuring device such as a graduate. The average of two tests shall be reported to the nearest cubic inch.

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EQUIPMENT FOR DETERMINING GELATION POINT OF HYDROCOLLOID IMPRESSION MATERIALS

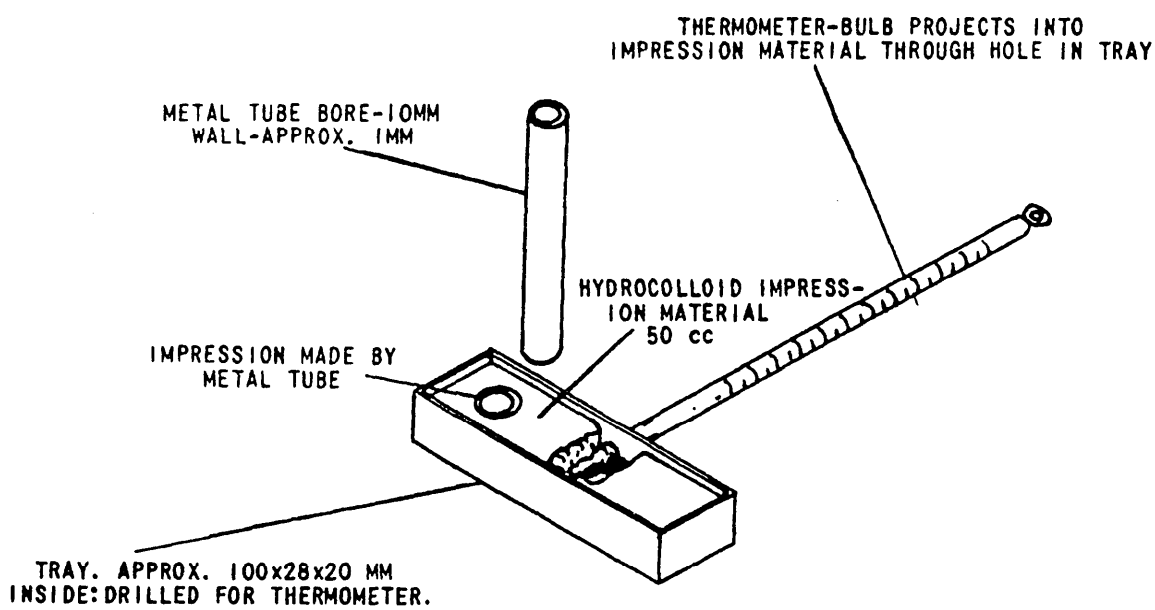


FIGURE 2

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Levels A or C as specified (see 6.2).

5.1.1 Unit of issue. One jar (1 gallon of impression material), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	4 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate container. Immediate container shall be a plastic, non-breakable, one-gallon jar with a screw-type lid.

5.1.3.2 Unit package. Each unit shall be packaged in a box of appropriate size and design, constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be Levels A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Immediate container. In addition to any commercial markings, each immediate container shall be marked to identify contents.

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

NOTE: Instructions for use may be printed on each immediate container or shall be inserted in each unit package.

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6. NOTES

6.1 Intended use. The duplicating material is intended for use with the monomer and polymer fluid resin procured under FSN 6520-125-9919 Denture Base Forming Kit., Fluid, Resin Technique (see 1.1, 3.3. and 4.4.1).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Literature reference. "Denture Bases Processed from a Fluid Resin", by Walter L. Shephard, Lieutenant Colonel, DC, USA, Walter Reed Army Medical Center, Washington, D.C., J. Pros. Dent., Vol. 19, No. 6, pages 561-572, June, 1968 (see 1.1, 3.2, 3.3 and 4.5.1).

6.4 Supersession data. This specification includes the requirements of Purchase Description No. 2, dated 16 March 1971.

6.5 Federal Stock Number (FSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog: 6520-133-1907 IMPRESSION MATERIAL, DENTAL, Duplicating Hydrocolloid, Agar Type.

6.6 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1317

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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MIL-I-36838

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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